PCB Pre-Fabrication Checklist

DIELCHA DIGILAB | Technical Reference & Engineering Quality Assurance

This checklist serves as the final quality gate before releasing Gerber, ODB++, drill files, and assembly documentation to PCB manufacturers. Complete all verification items to prevent costly respins and manufacturing delays.

1. Design Rule Checks (DRC) & Electrical Verification

Verification Item	Status
All electrical DRC errors and unconnected nets (airwires) resolved to zero.	Check
Clearance rules verified against manufacturer capabilities (trace/space, annular ring).	Check
Net connectivity verified against schematic; no dangling pins or floating nodes.	Check
Differential impedance tracks verified for correct trace width/gap spacing (if applicable).	Check

2. Silkscreen & Documentation Checks

Verification Item	Status
All component designators clearly visible and not overlapping pads, vias, or board edges.	Check
Polarity markings (Diodes, Electrolytic Capacitors, IC Pin 1) clearly indicated on silkscreen.	Check
Revision number, project name, company name, and date included on copper or silkscreen layer.	Check
Assembly drawing generated and matching current schematic and component references.	Check

3. Fabrication & Mechanical Layer Verification

Verification Item	Status
Board outline defined accurately on the mechanical/edge cuts layer with correct dimensions.	Check
Mounting hole diameters, locations, and copper clearance / keepout zones verified.	Check
Slot, cutout, and internal routing layers clearly documented with fabrication notes.	Check
Panelization and fiducial markers included for automated pick-and-place assembly (if required).	Check

4. Drill & Output Files Generation

Verification Item	Status
Excellon drill files generated with correct units, decimal format, and zero suppression.	Check
Gerber layers checked using a third-party Gerber viewer (GerberRS274X format verified).	Check
Bill of Materials (BOM) exported and cross-referenced with component designators.	Check

Pick-and-place (centroid) file exported with correct rotational orientation and coordinates.	Check